

# LPS Correctness and Validation: A Precise Experimental Plan

Turning the pre-submission audit roadmap into implementable, confound-controlled experiments

Pre-submission audit (companion to `lps_ps_lps_jasa_style_audit_2026-06-09.md`)

2026-06-09

## Abstract

This document specifies, for every item in the LPS roadmap (Tiers 0–4), an experiment or test that is detailed enough that two independent implementers would build the *same* experiment and reach the *same* verdict. Each entry states a falsifiable claim, the exact data-generating process (DGP), the exact `fit.lps` configuration, the statistic computed, a numeric acceptance criterion, and the validity safeguards that prevent the experiment from answering a confounded question. The shared conventions in Section 2 (RNG discipline, tolerances, the canonical DGP library, the linear-smoother extraction procedure and its leave-one-out caveat, the paired-comparison protocol) are defined once and referenced throughout; they are mandatory, not advisory. The priority order is correctness-first: Tier 0 establishes that the estimator is what it claims to be before any comparative or accuracy experiment is run.

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## 1 Scope and how to read this document

The object under test is the exported estimator `fit.lps` (and `predict.lps`) in package `geosmooth`, with the signature and defaults

```
fit.lps(X, y, foldid = NULL,
  support.grid = c(10L, 15L, 20L), degree.grid = 0:2,
  kernel.grid = c("gaussian", "tricube"),
  cv.folds = 5L, cv.seed = 1L, X.eval = NULL,
  coordinate.method = c("coordinates", "local.pca"),
  chart.dim = NULL,
  local.chart.method = c("pca", "second.order.svd"),
  auto.chart.support.metric = c("coordinates", "operator", "both"),
  auto.chart.selection.metric = c("coordinates", "operator"),
  backend = c("auto", "R", "cpp", "cpp.local.pca"),
  design.basis = c("orthogonal.polynomial.drop", "monomial",
    "weighted.qr", "weighted.qr.drop"),
  design.drop.tol = 1e-8,
  ridge.multiplier.grid = c(0, 1e-10, 1e-8),
  ridge.condition.max = 1e12,
  unstable.action = c("na", "mean"),
  outcome.family = c("gaussian", "bernoulli", "binomial"))
```

PS-LPS (`fit.ps.lps`) is deliberately out of scope here; this plan concerns LPS alone, per the stated priority of LPS algorithmic and implementation correctness.

Every experiment uses the same eight-field template: **Claim** (falsifiable); **Failure mode addressed**; **DGP**; **Configuration** (every non-default argument pinned); **Statistic**; **Acceptance criterion** (numeric); **Validity safeguards**; **Artifacts**. “Tests” (Tier 0, and the regression guards) are deterministic `testthat` assertions that must pass in CI. “Studies” (the comparative items in Tiers 1–4) are scripted experiments with predeclared decision rules; their outputs are reports plus a machine-readable verdict row.

## 2 Global conventions and shared infrastructure

### 2.1 Reproducibility and RNG discipline

1. Every stochastic quantity is generated immediately after an explicit `set.seed(s)`. Seeds are recorded in the result artifact. For replicated studies the replicate  $r \in \{1, \dots, R\}$  uses seed  $s_r = s_0 + r$  with a documented base  $s_0$ .
2. Cross-validation fold assignment is *always* supplied explicitly via the `foldid` argument; tests must not rely on the internal `cv.seed` default, because the internal fold construction is an implementation detail that may change. For paired comparisons the *same* `foldid` vector is passed to both arms.
3. Each result artifact stores `sessionInfo()`, the `geosmooth` package version and git commit hash, the BLAS/LAPACK identification, and the full argument list of every `fit.lps` call. A study whose artifact lacks these is invalid and must be re-run.
4. Determinism requirement: re-running any Tier 0 test twice in the same environment must yield bitwise-identical numeric results. Any test that is not deterministic is a defect in the test, not a tolerance to be widened.

### 2.2 Tolerance conventions

Let  $a$  be a computed value and  $b$  a reference. Define absolute error  $|a - b|$  and relative error  $|a - b|/(|b| + \delta)$  with floor  $\delta = 10^{-12}$ .

- **Algebraic identities** (e.g.  $\hat{y} = Sy$ , orthogonal vs. monomial span): absolute tolerance  $\tau_{\text{alg}} = 10^{-10}$ .
- **Polynomial reproduction / solver exactness**:  $10^{-8}$  for `design.basis`  $\in \{\text{orthogonal.polynomial.drop}, \text{weighted.qr}, \text{weighted.qr.drop}\}$ ; relaxed to  $10^{-6}$  for monomial on raw coordinates, where conditioning is worse *by design*. The relaxation must be justified in the test comment by reporting the realized design condition number.
- **Iterative solves** (local logistic IRLS): convergence tolerance  $10^{-7}$  on the coefficient step (matching the implementation), comparison tolerance  $10^{-6}$ .
- **Monte-Carlo quantities** (coverage, calibration, rates): no fixed tolerance; the acceptance criterion is a confidence interval or hypothesis test with the replicate count  $R$  chosen so the Monte-Carlo standard error is below one third of the decision threshold (state the power calculation in the script header).

### 2.3 Canonical data-generating processes (DGP library)

These are defined once. Every experiment references a DGP by tag and overrides only the parameters it names. All coordinates are double precision; all latent draws use Section 2.1.

#### G1 — ambient polynomial.

Ambient  $D \in \{2, 3\}$ .  $X_i \stackrel{iid}{\sim} \text{Unif}([-1, 1]^D)$ ,  $i = 1, \dots, n$ . Truth a fixed polynomial of degree  $p$ ; for  $D = 2, p = 2$  use  $f(x) = 0.5 + 1.0x_1 - 0.7x_2 + 0.4x_1^2 + 0.3x_1x_2 - 0.6x_2^2$ . Noiseless unless stated. Used for ambient reproduction and df checks.

**G2 — flat embedded subspace.**

Intrinsic  $d$ , ambient  $D > d$ . Latent  $u_i \sim \text{Unif}([-1, 1]^d)$ ; a fixed random orthonormal  $Q \in \mathbb{R}^{D \times d}$  (seed recorded);  $X_i = Qu_i$ . Truth a degree- $p$  polynomial in the *intrinsic*  $u$ . The embedding is a linear isometry, so a degree- $p$  local fit in the tangent chart reproduces  $f$  exactly. Used for intrinsic reproduction.

**G3a — paraboloid (known curvature).**

$d = 2, D = 3$ .  $u \sim \text{Unif}$  on the disk of radius  $\rho_0 = 1$ ;  $X = (u_1, u_2, (u_1^2 + u_2^2)/(2R))$ . Principal curvature  $\kappa = 1/R$  at the apex;  $R$  is the curvature knob. Truth options:  $f_{\text{lin}}(u) = u_1$  (intrinsic-linear, isolates curvature bias) and  $f_{\text{smooth}}(u) = \sin(\pi u_1) \cos(\pi u_2)$ .

**G3b — sphere cap (constant curvature).**

$d = 2, D = 3$ . A cap of a sphere of radius  $R$ : latent  $u \sim \text{Unif}$  on the planar disk  $\|u\| \leq \rho_0 < R$ , with  $X = (u_1, u_2, R - \sqrt{R^2 - \|u\|^2})$ . Constant Gaussian curvature  $1/R^2$  — a constant-positive-curvature complement to G3a’s position-varying paraboloid curvature. Truth options as in G3a.

**G3c — 1-D curve.**

$d = 1, D = 3$ . Helix  $X(t) = (\cos t, \sin t, ct)$ ,  $t \sim \text{Unif}[0, 2\pi)$ .

**G3d — torus patch.**

$d = 2, D = 3$ , standard torus with radii  $(R_1, R_2)$ , latent angles on a sub-rectangle to avoid wrap-around.

**G4 — stratified / varying dimension.**

Two glued strata sharing a boundary: stratum A is a 1-D segment  $\{(t, 0, 0) : t \in [-1, 0]\}$  (thickened by noise  $\eta$ ), stratum B is a 2-D patch  $\{(s_1, s_2, 0) : s_1 \in [0, 1], s_2 \in [-0.5, 0.5]\}$ . True local intrinsic dimension is 1 on A and 2 on B. Used to verify a dimension stabilizer does *not* blur the true boundary.

**G5 — clustered / repeated measures.**

$K$  clusters; centers  $c_k \sim \text{Unif}([-1, 1]^2)$ ; within cluster,  $m$  points  $x = c_k + \zeta$ ,  $\zeta \sim N(0, \sigma_x^2 I)$ . Response  $y = g(x) + b_k + \varepsilon$ ,  $b_k \sim N(0, \tau^2)$ ,  $\varepsilon \sim N(0, \sigma^2)$ , intra-class correlation  $\rho = \tau^2/(\tau^2 + \sigma^2)$ . Used for grouped-CV validity.

**G6 — binary surface.**

Smooth bounded log-odds  $\eta(x)$  with offset  $\alpha$  chosen so  $\mathbb{E}[p]$  hits a target prevalence;  $p(x) = \text{expit}(\alpha + \eta(x))$  clipped by construction to  $[0.05, 0.95]$  (no spike profiles);  $Y_i \sim \text{Bernoulli}(p(x_i))$ . Default  $\eta(x) = 1.5 \sin(\pi x_1)$  on  $x \in [-1, 1]^2$ .

**G7 — compositional / structural zeros.**

$X_i$  a  $D$ -part composition: draw Dirichlet( $\alpha$ ), set a documented subset of parts to exact 0, renormalize to the simplex. Used to characterize behavior on 16S-like inputs.

**2.4 Linear-smoother extraction, degrees of freedom, and the leave-one-out caveat**

Several experiments rely on the fact that, *for a single fixed configuration*  $\theta$  (one support size, one degree, one kernel, a fixed `chart.dim`, a fixed ridge), the LPS fitted vector is linear in the response:

$$\hat{y} = S(\theta) y, \quad S(\theta) \text{ depends only on } X, \text{ weights, } \theta, \text{ not on } y. \quad (1)$$

**Mandatory restriction.** Equation (1) holds only when no data-driven selection occurs. Therefore every experiment that uses  $S$  must call `fit.lps` with *singleton* grids (`support.grid`, `degree.grid`, `kernel.grid` of length one) and an explicit numeric `chart.dim` (never `"auto"/"local.auto"`, whose estimate, although  $y$ -independent, is a separate map). Using the CV-selected pipeline and then asserting linearity is a known trap: the selected  $\hat{\theta}(y)$  depends on  $y$ , so the end-to-end pipeline is *not* linear, and the assertion would (correctly) fail for the wrong reason.

**Extraction procedure.** With  $X.\text{eval} = X$ , fixed  $\theta$ , and the chosen `design.basis`: column  $j$  of  $S$  is  $S_{.j} = \text{fit}(e_j)$  where  $e_j$  is the  $j$ -th unit response vector (linearity makes finite differencing exact, so no step size is needed). Assemble  $S \in \mathbb{R}^{n \times n}$ , then  $\hat{y} = Sy$ , effective degrees of freedom  $\text{df} = \text{tr } S$ , and pointwise sensitivities  $S_{ii}$ .

## Two distinct leave-one-out notions (do not conflate).

1. *Response-removal LOO*: hold the geometry fixed, set  $y_i$ 's influence to zero. For a linear smoother this is *exactly*  $\hat{y}_{\text{resp}}^{(-i)}(x_i) = (\hat{y}_i - S_{ii}y_i)/(1 - S_{ii})$ .
2. *Training-removal LOO*: delete observation  $i$  from the training set and refit. For a  $k$ -NN-bandwidth smoother this *changes the support* of  $x_i$  (its  $k$ -th neighbor and hence the bandwidth  $h = \max$  distance change), so it is *not* equal to the response-removal shortcut.

The honest generalization target is training-removal LOO; the cheap shortcut is response-removal. Their gap is a real property of  $k$ -NN local regression and must be *measured*, not assumed away (see E0.3b). Asserting the  $(1 - S_{ii})$  identity against a training-removal refit is invalid and is explicitly prohibited.

## 2.5 Paired-comparison protocol

Any “method A vs. method B” accuracy comparison must hold fixed, across the two arms: the geometry  $X$ , the truth  $f$  (or  $p$ ), the realized response  $y$  (same noise draw), the `foldid`, and—unless the comparison is explicitly *about* that axis—the support size, degree, kernel, and `chart.dim`. The unit of analysis is the per-case delta  $\Delta = M_A - M_B$  of the target metric. Report the sign-test / Wilcoxon signed-rank across cases and replicates, the median  $\Delta$  with a bootstrap CI, and the full per-case table. A study that lets both arms re-select tuning by CV (so the arms differ on multiple axes at once) may be run only *in addition* to the matched comparison, never as its replacement.

## 2.6 Decision-rule conventions

Each study predeclares: the primary metric, the effect-size threshold that constitutes “materially better”, the replicate count  $R$  with its Monte-Carlo-error justification, and the statistical test. “Promote” / “do not promote” verdicts are emitted as a one-row machine-readable summary. Post-hoc threshold changes invalidate the study.

## 3 Tier 0 — Statistical-correctness battery (deterministic tests)

These are the cheapest possible proofs that the estimator is what it claims to be. They are `testthat` tests, must be deterministic (Section 2.1), and gate everything downstream. None of them involves model selection.

### 3.1 E0.1 — Polynomial reproduction (ambient and intrinsic)

**Claim.** A degree- $p$  LPS reproduces any global polynomial of degree  $\leq p$  with error at machine-solve precision, for every kernel and every `design.basis`, and—on a *flat* manifold—for the `local.pca` chart at the correct intrinsic dimension.

**Failure mode addressed.** Centering, design-matrix construction, or chart-coordinate bugs that introduce bias even on data the model can represent exactly. This is the single most diagnostic LPS test and is currently absent.

**DGP.** (a) Ambient: G1 with  $D \in \{2, 3\}$ ,  $p \in \{1, 2\}$ ,  $n = 200$ , noiseless. (b) Intrinsic: G2 with  $d \in \{1, 2\}$ ,  $D = d + 2$ ,  $p \in \{1, 2\}$ ,  $n = 200$ , noiseless.

**Configuration.** Singleton grids; support large enough for full rank.

```
# (a) ambient reproduction
fit.lps(X, y, foldid = rep(1:2, length.out = n),
        support.grid = K, degree.grid = p, kernel.grid = ker,
        coordinate.method = "coordinates", backend = "R",
        design.basis = basis, ridge.multiplier.grid = 0,
        ridge.condition.max = Inf, unstable.action = "na")
# (b) intrinsic reproduction: coordinate.method = "local.pca", chart.dim = d
```

Sweep `ker` over all four kernels, `basis` over all four design bases, with  $K = 3 \cdot c_p$  where  $c_p$  is the design-column count  $c_p = 1 + d + \binom{d+1}{2} \mathbf{1}\{p = 2\}$  (ambient uses  $d = D$ ). The factor 3 guarantees the support is over-determined for the degree.

**Statistic.**  $\max_i |\hat{f}(x_i) - f(x_i)|$  over  $X.\text{eval} = X$ .

**Acceptance criterion.**  $< 10^{-8}$  for orthogonal/QR bases,  $< 10^{-6}$  for `monomial` (Section 2.2), for *all* kernel  $\times$  basis  $\times$  degree combinations. The test also asserts the reported minimum design rank equals  $c_p$ .

**Validity safeguards.** (i) The intrinsic case uses a *flat* (isometric) embedding only: on a curved manifold exact reproduction is false (curvature bias), so using G3 here would be a spurious failure. (ii)  $X$  drawn from a continuous distribution with a fixed seed so supports are in general position (no accidental collinearity); the rank assertion guards this. (iii) `chart.dim` set to the true  $d$ , never "auto", so the test isolates reproduction from dimension estimation.

**Artifacts.** `testthat` file `test-lps-reproduction.R`; on failure, dump the offending neighborhood's design, weights, and condition number.

### 3.2 E0.2 — Linear-smoother identity and degrees of freedom

**Claim.** For a fixed configuration, (i)  $\hat{y} = Sy$  exactly; (ii)  $S$  is independent of  $y$ ; (iii)  $\text{df} = \text{tr } S$  equals the finite-difference trace  $\sum_i \partial \hat{y}_i / \partial y_i$ .

**Failure mode addressed.** Any hidden  $y$ -dependence in the weighting or support (which would break every downstream  $\text{df}$  / variance / CV-shortcut claim), and an incorrect  $\text{df}$  accounting.

**DGP.** G1,  $D = 2$ ,  $n = 120$ ,  $y$  realized once with  $N(0, 1)$  entries (the values are irrelevant; only linearity is tested).

**Configuration.** Singleton grids; `coordinate.method` both "coordinates" and "local.pca" with explicit `chart.dim=2`; `design.basis = "orthogonal.polynomial.drop"`; fixed ridge.

**Statistic.** Build  $S$  per Section 2.4. Compute (i)  $\max |\mathbf{fit}(y) - Sy|$  for three independent random  $y$ ; (ii)  $\max |S^{(y_a)} - S^{(y_b)}|$  for two different base responses; (iii)  $|\text{tr } S - \sum_i (\mathbf{fit}(y + \epsilon e_i)_i - \mathbf{fit}(y)_i) / \epsilon|$  with  $\epsilon = 1$  (exact by linearity).

**Acceptance criterion.** (i)  $< \tau_{\text{alg}} = 10^{-10}$ ; (ii)  $< 10^{-12}$  (literally the same matrix); (iii)  $< 10^{-10}$ .

**Validity safeguards.** Singleton grids and numeric `chart.dim` per Section 2.4; assert that the run performed no candidate selection (CV table has one row).

**Artifacts.** `test-lps-linear-smoother.R`; persist one  $S$  matrix as a fixture for E0.3 and E4.

### 3.3 E0.3a — Response-removal LOO shortcut (algebraic)

**Claim.** Whatever leave-one-out / GCV residual the LPS (or GCV) code reports matches the analytic shortcut  $(y_i - \hat{y}_i) / (1 - S_{ii})$  formed from the *independently* extracted  $S$  of E0.2. This is a unit test of the **GCV/LOO formula implementation**, not a validation of the estimator: response-removal LOO equals that shortcut *by derivation*, so it cannot test anything beyond whether the code computes the formula correctly. The scientifically meaningful leave-one-out question — training-removal, where the  $k$ -NN support changes — is E0.3b, which carries the weight.

**Failure mode addressed.** An incorrect GCV/LOO algebra in the code that would silently bias any  $S$ -based selection.

**DGP / Configuration.** As E0.2.

**Statistic.** Compare the code-reported per-point LOO/GCV residual to  $(y_i - \hat{y}_i) / (1 - S_{ii})$  with  $S$  from E0.2. If the package has no separate GCV/LOO path for LPS, this test has no independent target: it is then deferred (folded into E0.2) and must *not* be reconstructed purely from  $S$  and then “verified” against itself — that would be tautological.

**Acceptance criterion.**  $\max_i$  relative error  $< 10^{-8}$  against the independent code path (or deferred if no such path exists); rows with  $S_{ii} \geq 1 - 10^{-6}$  are reported and excluded (near-interpolation).

**Validity safeguards.** Do *not* compare against a training-removal refit here (Section 2.4).

### 3.4 E0.3b — Training-removal vs. shortcut gap (characterization study)

**Claim (characterization, not pass/fail).** Quantify how far the cheap response-removal shortcut is from honest training-removal LOO for  $k$ -NN-bandwidth LPS, as a function of  $k$ .

**Failure mode addressed.** Silently using the  $(1 - S_{ii})$  shortcut as if it were honest CV when the  $k$ -NN support change makes it optimistic.

**DGP.** G3a ( $f_{\text{smooth}}$ ),  $n = 400$ , noise  $\sigma = 0.05$ ;  $k \in \{8, 15, 30, 60\}$ ;  $R = 20$  replicates.

**Configuration.** Fixed degree 1, `chart.dim=2`, `gaussian`; training-removal LOO for a random subsample of 200 indices via re-fitting on  $X[-i]$  and predicting  $x_i$ .

**Statistic.** Per  $i$ :  $g_i = \hat{y}_{\text{train}}^{(-i)}(x_i) - \hat{y}_{\text{resp}}^{(-i)}(x_i)$ ; report median and 90th percentile of  $|g_i|/\text{sd}(y)$  by  $k$ .

**Acceptance criterion (advisory threshold).** If median relative gap  $< 0.05$  for the  $k$  used in practice, the shortcut may stand in for honest CV in reports; otherwise reports must use training-removal CV (or grouped CV, E1.10). The verdict is recorded, not asserted.

**Validity safeguards.** Same noise draw across the two LOO notions; exclude near-interpolation rows; hold  $k$  fixed within a cell.

### 3.5 E0.4 — Boundary-bias property (local linear vs. local constant)

**Claim.** Local linear (degree 1) is first-order boundary-bias-free whereas local constant (degree 0) carries  $O(h)$  boundary bias: near the domain edge the degree-1 error is a small multiple of its interior error, while the degree-0 boundary/interior error ratio is large.

**Failure mode addressed.** A degree-1 implementation that has lost the boundary-correction property (e.g. through mis-centering).

**DGP.** 1-D design  $x_i$  on  $[0, 1]$ ,  $n = 300$  (equally spaced plus a fixed jitter, seed recorded); truth  $f(x) = e^x$  (nonzero slope and curvature at the boundary); noiseless.

**Configuration.** `coordinate.method="coordinates"`, singleton  $K = 30$ , `gaussian`, degrees 0 and 1 in separate fits, `ridge.multiplier.grid=0`.

**Statistic.** Boundary set  $B = \{x_i : x_i < h \text{ or } x_i > 1 - h\}$  with  $h$  the realized  $K$ -NN bandwidth at the edge; interior set  $I$  the complement. Compute  $\bar{e}_B^{(p)} = \text{mean}_B|\hat{f} - f|$  and  $\bar{e}_I^{(p)}$  for  $p \in \{0, 1\}$ .

**Acceptance criterion.**  $\bar{e}_B^{(1)}/\bar{e}_B^{(0)} < 0.5$  and  $\bar{e}_B^{(1)}/\bar{e}_I^{(1)} < 3$  and  $\bar{e}_B^{(0)}/\bar{e}_I^{(0)} > 5$ .

**Validity safeguards.** Noiseless (isolates bias); same  $x$ ,  $h$ , kernel across the degree arms (paired); truth chosen nonlinear so degree 1 does not reproduce it exactly (which would make all errors 0 and the ratio undefined).

### 3.6 E0.5 — Consistency and rate (mean-response)

**Claim.** With the bandwidth schedule below, Truth-RMSE decreases at the local-linear manifold rate  $\text{RMSE} \asymp n^{-2/(d+4)}$  on a curved  $d$ -manifold.



**Failure mode addressed.** A bias floor masquerading as inconsistency (the trap of fixing  $k$  so bias never vanishes), or an estimator that fails to converge.

**DGP.** G3a ( $f_{\text{smooth}}, d = 2$ ), noise  $\sigma = 0.1$ ;  $n \in \{200, 400, 800, 1600, 3200\}$ ;  $R = 30$  replicates per  $n$ .

**Configuration.** Fixed degree 1, `chart.dim=2, gaussian, design.basis="orthogonal.polynomial.drop"`. Support grows as  $k(n) = \text{round}(cn^{4/(d+4)})$  (for  $d = 2$ :  $k \propto n^{2/3}$ ), with  $c$  fixed so  $k(200) = 20$ ; `support.grid=k(n)` (singleton, so no selection).

**Statistic.**  $\text{RMSE}_p(n) = \left(\frac{1}{n} \sum_i (\hat{f}(x_i) - f(x_i))^2\right)^{1/2}$ , averaged over replicates; fit log RMSE =  $a + b \log n$ .

**Acceptance criterion.** (Smoke, required) the 95% CI for  $b$  lies entirely below  $-0.1$ . (Rate, recommended) the CI for  $b$  contains the theoretical  $-2/(d+4) = -1/3$ , or the deviation is explained by a stated finite-sample/boundary argument.

**Validity safeguards.** Bandwidth schedule pinned to the optimal growth (otherwise a bias floor invalidates the rate); same seed structure across  $n$ ; report bias and variance components separately so a plateau is correctly attributed.

### 3.7 E0.6 — Binary recovery and calibration

**Claim.** Both binary modes produce probability estimates that (i) are consistent ( $\text{RMSE}_p \downarrow$  with  $n$ ) and (ii) are calibrated at large  $n$  (calibration slope  $\approx 1$ , intercept  $\approx 0$ ).

**Failure mode addressed.** Mis-scaled or mis-clipped probabilities; a selection metric that does not target probability quality; silent degradation under class imbalance.

**DGP.** G6 on  $x \in [-1, 1]^2$  (treated as `coordinates`), prevalence  $\mathbb{E}[p] \in \{0.1, 0.3, 0.5\}$ ,  $n \in \{500, 1000, 2000, 4000\}$ ,  $R = 40$ .

**Configuration.** `outcome.family`  $\in \{\text{"bernoulli"}, \text{"binomial"}\}$ , `support.grid/degree.grid/kernel.grid` as the routine grids (selection *is* allowed here—this is an accuracy study, not an  $S$  study), `backend="R"`.

**Statistic.**  $\text{RMSE}_p$  vs. the known  $p$ ; reliability via 10 equal-count bins of  $\hat{p}$  (observed frequency vs. mean  $\hat{p}$ ); calibration slope/intercept from the logistic regression  $Y \sim \text{logit}(\hat{p})$  on a held-out half.

**Acceptance criterion.** (i) the log  $\text{RMSE}_p$  vs. log  $n$  slope CI lies below  $-0.1$  at every prevalence; (ii) at  $n = 4000$ , calibration slope  $\in [0.8, 1.25]$  and intercept  $\in [-0.25, 0.25]$ . The Bernoulli-vs-binomial difference is *reported*, not asserted.

**Validity safeguards.**  $p$  bounded in  $[0.05, 0.95]$  (no spike profiles); calibration fit on data disjoint from the LPS fit; at prevalence 0.1 with  $n = 500$  expect logistic fallbacks—stratify all metrics by `logistic.diagnostics` fallback fraction and report it.

### 3.8 E0.7 — Cross-validation has no response leakage

**Claim.** Perturbing a held-out response  $y_i$  leaves  $i$ ’s own out-of-fold prediction exactly unchanged.

**Failure mode addressed.** Any leakage of held-out responses into their own out-of-fold predictions (which would make CV optimistic).

**DGP.** G1,  $D = 2$ ,  $n = 100$ , explicit `foldid` with 5 folds.

**Configuration.** Fixed configuration (singleton grids) so no selection confounds the test. Out-of-fold prediction of fold  $f$  obtained by `fit.lps` on  $\{j : \text{foldid}_j \neq f\}$  with  $X.\text{eval} = \{j : \text{foldid}_j = f\}$ .

**Statistic.** For a held-out index  $i$  (in fold  $f$ ): refit after replacing  $y_i$  by  $y_i + \Delta$  ( $\Delta = 10$ ) and recompute the out-of-fold prediction at  $i$ . Report  $|\hat{y}_i^{\text{out}}(\text{perturbed}) - \hat{y}_i^{\text{out}}(\text{original})|$ .

**Acceptance criterion.**  $< 10^{-12}$  (exactly zero up to roundoff). As a positive control, perturbing  $y_i$  *does* change the prediction of some  $j$  in another fold that has  $i$  in its training support (assert change  $> 0$ ), proving the test is not vacuous.

**Validity safeguards.** Fixed config; the geometry (and any `auto` dimension) is  $y$ -independent, so the test isolates response leakage.

### 3.9 E0.8 — Degenerate-geometry pathologies

**Claim.** On each pathology, LPS terminates, returns finite predictions where defined, and exhibits the specified guarded behavior (no silent mean fallback when `unstable.action="na"`; correct diagnostics).

**Failure mode addressed.** Crashes, silent fallbacks, or NaNs on inputs that real omics data routinely produce.

**DGP / cases (each a separate test).**

1. *Duplicate points*: replicate a single  $x$  value  $k$  times in a support. Expect finite prediction; a fully-degenerate support collapses to the weighted mean (degree 0 behavior) and the rank diagnostic equals 1.
2. *Collinear support*: points on a line in  $\mathbb{R}^2$  with degree 2. Expect rank-deficient quadratic dropped by `orthogonal.polynomial.drop`; with `unstable.action="na"` a configuration that cannot be solved yields NA (and is thus avoided by selection), never a silent mean.
3. *Constant response*:  $y \equiv c$ . Expect  $\hat{f} \equiv c$  within  $10^{-10}$  for all degrees, kernels, bases.
4. *Distance ties*: symmetric  $X$  with exact ties. Expect deterministic support choice (index tie-break) and bitwise-identical results across repeated runs.
5. *Class imbalance (binary)*: G6 at prevalence 0.02,  $n = 300$ , `outcome.family="binomial"`. Expect no crash; `logistic.diagnostics` reports the event-rate fallback count; fitted probabilities  $\in [0, 1]$ ; selection either returns a finite log-loss or fails fast with the documented “no finite selection score” error.
6. *Compositional / structural zeros*: G7,  $D = 6$ . Expect finite predictions; record the fraction of supports with zero bandwidth ( $h = 0$ ) and the fallback handling.

**Acceptance criterion.** Per case as stated; all assertions deterministic.

**Validity safeguards.** Test both `unstable.action` settings where relevant so the guarded-vs-legacy behavior is pinned; assert on diagnostics, not only on non-crashing.

**Artifacts.** `test-lps-degenerate.R`; one fixture per case.

## 4 Tier 1 — Honest selection and bandwidth

### 4.1 E1.9 — Decouple bandwidth from support size

**Claim.** (a) The current kernel weighting ties the effective bandwidth to the  $K$ -NN radius, so “gaussian” is near-flat (effective sample size  $\approx K$ ) and compact kernels zero the  $K$ -th neighbor (effective support  $K - 1$ ); (b) a bandwidth multiplier  $b$  that sets  $h = b \cdot (K\text{-th NN distance})$  reproduces current behavior exactly at  $b = 1$ ; (c) adding  $b$  to the selection grid does not worsen, and on curved truths improves, Truth-RMSE.

**Failure mode addressed.** Kernel selection silently selecting an effective-sample-size artifact; an unexposed bandwidth that prevents reaching the curvature-relevant scale.

**DGP.** (a)/(b) deterministic on a single fixed support (any G1 neighborhood). (c) G3a and G3d,  $n = 600$ ,  $\sigma \in \{0.03, 0.1\}$ ,  $R = 30$ , paired (Section 2.5).

#### **Configuration / statistic.**

1. Characterization (regression test): call the internal weight function on a fixed distance vector for each kernel; compute Kish effective sample size  $ESS = (\sum w)^2 / \sum w^2$  and the  $K$ -th weight. Assert  $ESS/K > 0.9$  for `gaussian`,  $ESS/K < 0.85$  for `tricube`, and  $w_{(K)} / \max_j w_j < 10^{-6}$  for all compact kernels. This *pins current behavior* so a future change is visible.
2. Backward-compatibility (after implementing  $b$ ): assert fits with  $b = 1$  equal the current fits within  $\tau_{\text{alg}}$ .
3. Benefit study: arms = {select over  $(K)$  only} vs. {select over  $(K, b)$ ,  $b \in \{0.5, 1, 2, 4\}$ }; metric Truth-RMSE; predeclared “material” = median  $\Delta > 5\%$  with Wilcoxon  $p < 0.05$ .

**Acceptance criterion.** (a) assertions hold; (b)  $b = 1$  exactness; (c) verdict recorded; promotion only if the material threshold is met and runtime cost is reported.

**Validity safeguards.** The benefit study is matched on  $(K, \text{degree}, \text{kernel}, \text{chart.dim})$  except the bandwidth axis; same noise draw; the characterization uses the *actual* internal weighting routine, not a re-implementation.

### 4.2 E1.10 — Nested and grouped cross-validation

**Claim.** (a) The selected-minimum CV score is optimistic; nested CV corrects it and tracks held-out test error. (b) Random  $K$ -fold underestimates error under cluster dependence; leave-cluster-out CV closes the gap.

**Failure mode addressed.** Reporting a selection-optimistic number on real data; CV leakage through dependent (subject/replicate) samples.

**DGP.** (a) G3a,  $n = 800$  train +  $n_{\text{test}} = 4000$  independent test,  $R = 40$ . (b) G5 with  $K = 40$  clusters,  $m = 20$ ,  $\rho \in \{0.3, 0.6\}$ ; an independent test set of fresh clusters;  $R = 40$ .

**Configuration.** Routine grids (selection is the object of study). Outer loop: 5 folds; inner loop: 5 folds for selection. For (b) compare `foldid` built by random assignment vs. by cluster id.

**Statistic.** (a)  $|\widehat{\text{RMSE}}_{\bullet} - \text{RMSE}_{\text{test}}| / \text{RMSE}_{\text{test}}$  for  $\bullet \in \{\text{selected-min, nested}\}$ . (b) the same for random-fold and cluster-fold estimates against fresh-cluster test error.

**Acceptance criterion.** (a) nested relative error  $< 0.10$  *and* nested  $\geq$  selected-min in expectation (optimism sign correct). (b) random-fold relative error exceeds cluster-fold relative error by a predeclared margin ( $> 0.10$  at  $\rho = 0.6$ ); cluster-fold within 0.10 of truth.

**Validity safeguards.** Test clusters disjoint from training clusters (no cluster appears in both); identical truth across arms; report the realized  $\rho$ .

### 4.3 E1.11 — Stabilizing the local intrinsic-dimension estimate

**Claim.** (a) Per-anchor dimension estimates from small supports are high-variance; (b) shrinkage-to-global and graph-smoothing of the dimension field reduce misclassification *without* blurring a true dimension boundary and without worsening Truth-RMSE; (c) a points-per-parameter gate prevents pathological degree-2-in-high-dimension selections.

**Failure mode addressed.** The documented harm from dimension over-estimation (the FB01 finding); a stabilizer that over-smooths across genuine dimension changes.

**DGP.** (a)/(b) homogeneous: G3a ( $d = 2$ ),  $n \in \{200, 800\}$ ,  $K \in \{10, 20, 40\}$ ,  $R = 50$ . (b) boundary safeguard: G4 (true dim 1 on stratum A, 2 on B). (c) any G2 with  $d = 3$ ,  $D = 5$ .

**Configuration.** `coordinate.method="local.pca", chart.dim="local.auto"; record chart.dim.by.eval.` Stabilizers implemented as post-processors of the per-anchor estimate: shrink-to-global (mode/median of the field) and  $k$ -NN majority vote.

**Statistic.** (a) misclassification rate  $\frac{1}{n} \sum_i \mathbf{1}\{\hat{d}_i \neq d\}$  and its across-replicate variance. (b) on G4, the misclassification rate *within* each stratum and at anchors whose support straddles the boundary. (c) whether the gated selector ever returns degree 2 with  $K < c \cdot c_2$  (set  $c = 1.5$ ).

**Acceptance criterion.** (a) report the variance decay with  $K$  (characterization). (b) a stabilizer is acceptable only if it reduces homogeneous-region misclassification by a predeclared margin ( $> 30\%$  at  $K = 10$ ) *and* does not increase boundary-straddling misclassification on G4 by more than 5 percentage points *and* is non-inferior on Truth-RMSE (CI of  $\Delta$  excludes a  $> 2\%$  degradation). (c) with the gate, the offending candidates receive a non-finite/penalized score; without it, they can be selected (positive control).

**Validity safeguards.** The G4 boundary test is mandatory: a stabilizer that looks good only on homogeneous data but smears the stratified boundary is rejected. Dimension truth on G4 is defined per stratum and recorded with the geometry.

## 5 Tier 2 — Binary path and numerical hygiene

### 5.1 E2.12 — Binary selection metric consistency and log-loss clipping

**Claim.** (a) Bernoulli-mode selection currently scores *unclipped* predictions while deployment is clipped, and this can change the selected candidate; the fix is to score the deployed (clipped) metric. (b) A log-loss truncation of  $10^{-15}$  makes the selection score unstable;  $10^{-6}$  is stable.

**Failure mode addressed.** Train/deploy metric mismatch; a high-variance selection criterion dominated by a single confident error.

**DGP.** (a) a constructed case: G6 with a sharp region forcing some raw predictions outside  $[0, 1]$ ,  $n = 400$ , where clipped vs. unclipped Brier rank two candidates differently. (b) G6 with one deliberately confident-wrong held-out point.

**Statistic.** (a) selected candidate under clipped vs. unclipped Brier. (b) selected candidate and log-loss value under clip  $\in \{10^{-15}, 10^{-6}, 10^{-3}\}$ .

**Acceptance criterion.** (a) a regression test asserting the selection score equals the deployed (clipped) metric after the fix; the pre-fix discrepancy is demonstrated in the same file as a documented motivating case. (b) the *score* at clip  $10^{-15}$  is demonstrably dominated by a single confident-wrong point (the actual defect); adopt and pin a documented finite clip ( $10^{-6}$ ). Selection *stability* across  $\{10^{-6}, 10^{-3}\}$  is reported as a diagnostic, **not gated** — near-ties may legitimately reselect, so cross-clip invariance is a robustness goal, not a hard correctness criterion.

**Validity safeguards.** Construct the (a) case so the ranking flip is real (verify by hand), not a tie; hold all else fixed.

### 5.2 E2.13 — Ridge penalty structure alignment

**Claim.** In the orthogonal basis the ridge currently penalizes the constant direction, so as the ridge grows the prediction shrinks toward 0 rather than toward the local weighted mean; after aligning the penalty (leave the constant direction unpenalized), large ridge shrinks toward the local weighted mean, and tiny ridge is prediction-invariant.

**Failure mode addressed.** A statistically wrong shrinkage target hidden behind currently-tiny ridge multipliers.

**DGP.** G1,  $D = 2$ ,  $n = 150$ , mild noise.

**Configuration.** Fixed config; `design.basis="orthogonal.polynomial.drop"`; sweep `ridge.multiplier.grid` as singletons  $\rho \in \{0, 10^{-8}, 10^{-2}, 1, 10^2\}$  (with `ridge.condition.max=Inf` so the chosen  $\rho$  is actually applied).

**Statistic.** For each anchor,  $\hat{f}_\rho(x_i)$ ; compare  $\hat{f}_{\rho \rightarrow \infty}$  to the local weighted mean  $\bar{y}_i^w$  and to 0.

**Acceptance criterion.** After the fix:  $|\hat{f}_{\rho=10^2} - \bar{y}^w|$  small relative to  $|\hat{f}_{\rho=10^2} - 0|$  (shrinks to the local mean, not to zero); and  $|\hat{f}_{\rho=10^{-8}} - \hat{f}_{\rho=0}| < 10^{-6}$  (tiny-ridge invariance). A pre-fix test documents the shrink-to-zero behavior so the change is intentional and visible.

**Validity safeguards.** `ridge.condition.max=Inf` so the guard does not silently skip the large- $\rho$  candidate; paired across  $\rho$  on the same data.

### 5.3 E2.14 — Local logistic robustness (separation)

**Claim.** The local logistic IRLS is monotone and bounded under near-separation: with step-halving the deviance is non-increasing across iterations, coefficients and fitted probabilities stay finite and in range, and non-convergence triggers a deterministic, telemetered fallback.

**Failure mode addressed.** IRLS divergence/oscillation under separation, NaNs, or a “win” that is actually a mass of event-rate fallbacks.

**DGP.** A single near-separable support:  $z \in \mathbb{R}$  with  $y = \mathbf{1}\{z > 0\}$  plus one flipped label to avoid exact separation; weights from `gaussian`.

**Configuration.** Call the local logistic solver (degree 1, `outcome.family="binomial"`, `backend="R"`) directly on the constructed support; record per-iteration deviance.

**Statistic.** Per-iteration deviance sequence; final coefficients; fitted probability; telemetry (`converged`, `fallback.path`, `event.rate.fallback`).

**Acceptance criterion.** With step-halving, deviance is non-increasing to within  $10^{-8}$  per step; final  $\hat{p} \in (0, 1)$  and  $|\hat{\beta}| < \infty$ ; if the convergence tolerance is not met within the iteration cap, the documented fallback fires and is recorded. Exact separation (no flipped label) must hit the fallback, not loop or NaN.

**Validity safeguards.** Test both the near-separable and exactly-separable cases; assert on the deviance *trajectory*, not only the endpoint, so oscillation is caught.

## 6 Tier 3 — Curvature-chart re-evaluation, done correctly

The H4/H5 “no gains” result for `second.order.svd` is confounded: both arms selected degree 2 (the diagnostics’ own top predictor of the effect is the cross-arm degree difference), the error budget was noise-dominated ( $\sigma \approx \text{Truth-RMSE}$ ), curvature was estimated from  $\leq 18$ -point supports, and the metric averaged over the whole manifold. A degree-2 polynomial in tangent coordinates already absorbs much of the leading curvature signal, so a degree-2-vs-degree-2 comparison cannot isolate the chart’s contribution. The two are not identical mechanisms — a degree-2 tangent polynomial adds function-side flexibility, whereas the second-order chart corrects the coordinate map itself — but they overlap heavily on the leading bias term, which is enough to make the old test uninformative. The following isolates the curvature chart’s value.

## 6.1 E3.1 — Curvature-chart benefit under a degree-1, low-noise, curvature-stratified design

**Claim.** On genuinely curved manifolds, at *degree* 1 and adequate support, the `second.order.svd` chart reduces Truth-RMSE in high-curvature regions relative to the `pca` chart; and degree-1+`second.order.svd` approaches degree-2+`pca` (confirming that degree-2 tangent polynomials make the chart correction *largely redundant in this regime* — not that the two are identical mechanisms).

**Failure mode addressed.** A confounded comparison (degree-2 both arms; noise-dominated; whole-manifold averaging) that cannot reveal a curvature effect even if present.

**DGP.** G3a, G3b (sphere cap), G3d (torus patch); curvature knob giving a low- and a high-curvature setting per family;  $n \in \{400, 1600\}$ ; noise  $\sigma \in \{0.01, 0.03\}$  (low, so bias is visible);  $R = 30$ ; plus the VALENCIA-derived real-geometry probe as an external check.

**Configuration.** *Matched* arms differing only in `local.chart.method`  $\in \{\text{"pca"}, \text{"second.order.svd"}\}$ . Fix `degree` = 1, `chart.dim` = true  $d$ , the same kernel, and support  $K \geq 3 \cdot q_2$  where  $q_2 = d + \binom{d+1}{2}$  is the curvature parameter count (for  $d = 2$ ,  $q_2 = 5$ , so  $K \geq 20$ ; sweep  $K \in \{25, 40, 60\}$ ). A third reference arm: degree 2 + `pca`, same  $K$ . Paired (Section 2.5).

**Statistic.** Local curvature magnitude  $\kappa(x)$  is known analytically; bin eval points into curvature deciles. Primary metric: Truth-RMSE within the top curvature decile,  $\text{RMSE}^{\text{hi}}$ . Secondary: whole-manifold Truth-RMSE; and  $\text{RMSE}(\text{deg1} + 2\text{nd}) - \text{RMSE}(\text{deg2} + \text{pca})$ .

**Acceptance / promotion criterion (predeclared).** Promote the curvature chart beyond “experimental” only if median over cases of  $\Delta^{\text{hi}} = \text{RMSE}_{\text{pca}}^{\text{hi}} - \text{RMSE}_{2\text{nd}}^{\text{hi}}$  is  $> 10\%$  relative with a sign-test  $p < 0.05$  across cases  $\times$  replicates, *and* runtime ratio is reported, *and* the fallback rate (chart dim = ambient, rank deficiency) is  $< 10\%$  in the promoted regime. Separately, report whether degree-1+2nd  $\approx$  degree-2+pca (redundancy confirmation).

**Validity safeguards.** Degree fixed at 1 in both compared arms (the central fix); support sized for stable curvature estimation; low noise so bias dominates; curvature-stratified metric so localized gains are not averaged away; matched seeds/support/kernel/dim; fallback accounting mandatory; flat datasets excluded.

## 6.2 E3.2 — Curvature-bias unit test (theory-anchored)

**Claim.** For an intrinsic-linear truth on a known-curvature manifold, degree-1+`pca` exhibits the analytically predicted  $O(\kappa h^2)$  bias, and degree-1+`second.order.svd` removes its leading term.

**Failure mode addressed.** A curvature-correction implementation that is plausible but does not actually cancel the known curvature term (a correctness question independent of accuracy gains).

**DGP.** G3a with  $f_{\text{lin}}(u) = u_1$  (intrinsic-linear), noiseless; curvature  $\kappa = 1/R$  swept over  $R \in \{1, 2, 4, 8\}$ ; support radius held fixed so  $h$  is comparable across  $R$ .

**Statistic.** At anchors in the high-curvature region, bias  $\beta^{(p,\text{chart})}(x) = \hat{f}(x) - f(x)$ ; fit  $|\bar{\beta}_{\text{pca}}|$  vs.  $\kappa h^2$  (expect linear through the origin); ratio  $|\bar{\beta}_{2\text{nd}}|/|\bar{\beta}_{\text{pca}}|$ .

**Acceptance criterion.**  $|\bar{\beta}_{\text{pca}}|$  vs.  $\kappa h^2$  has slope significantly  $> 0$  and  $R^2 > 0.9$  (the predicted scaling), and  $|\bar{\beta}_{2\text{nd}}|/|\bar{\beta}_{\text{pca}}| < 0.3$  at the highest curvature (leading term removed). If the chart does *not* reduce the bias here, the correction is wrong regardless of E3.1.

**Validity safeguards.** Noiseless (pure bias); intrinsic-linear truth so degree-1 would be exact *but* for curvature, isolating the curvature term;  $h$  held fixed across  $R$  so the  $\kappa h^2$  scaling is identifiable.

## 7 Tier 4 — Uncertainty quantification

### 7.1 E4.1 — Pointwise variance and confidence-band coverage

**Claim.** The linear-smoother variance  $\text{Var}(\hat{y}_i) = \sigma^2 \sum_j S_{ij}^2$  yields pointwise bands whose empirical coverage matches nominal in the interior; coverage degrades where bias dominates (boundary/high curvature), as expected, and this is reported rather than masked.

**Failure mode addressed.** Shipping point estimates with no uncertainty; or claiming nominal coverage globally where bias guarantees undercoverage.

**DGP.** G3a ( $f_{\text{smooth}}$ ), homoskedastic  $\varepsilon \sim N(0, \sigma^2)$  with  $\sigma = 0.1$  *known*;  $n = 1200$ ;  $R = 500$  replicates (so coverage Monte-Carlo SE  $\approx \sqrt{0.95 \cdot 0.05/500} \approx 0.01$ ).

**Configuration.** Fixed config (singleton grids), `chart.dim=2`, degree 1. Build  $S$  (Section 2.4); band  $\hat{y}_i \pm z_{0.975} \hat{\sigma} \|S_{i\cdot}\|_2$  with  $\hat{\sigma}^2 = \text{RSS}/(n - \text{tr } S)$ .

**Statistic.** Per eval point, empirical coverage of  $f(x_i)$  across replicates; stratify into interior, boundary ( $x$  within  $h$  of the disk edge), and top-curvature-decile sets.

**Acceptance criterion.** Interior average coverage  $\in [0.93, 0.97]$  (using known  $\sigma$ ); with the plug-in  $\hat{\sigma}$ ,  $\in [0.92, 0.98]$ . Boundary/high-curvature undercoverage is reported with its magnitude; if a bias-corrected band is implemented, re-test that its interior+boundary coverage both fall in range.

**Validity safeguards.** Known  $\sigma$  first (separates variance estimation from band construction); honest stratification—do *not* average interior and boundary coverage into one headline number, since bias-driven boundary undercoverage is real and must be visible.

## 8 Execution order and dependency map

1. **Tier 0 first, as CI gates.** E0.2 (linear smoother) and E0.1 (reproduction) are the keystones: E0.3a, E0.5, E4.1 reuse the  $S$  machinery from E0.2, and any failure of E0.1/E0.2 invalidates everything downstream, so they run before any study.
2. **Tier 1 next**, because honest selection and bandwidth semantics underlie every accuracy number reported in Tiers 3–4. E1.10 (nested/grouped CV) should be in place before any real-data Truth-unknown claim.



3. **Tier 2** can proceed in parallel with Tier 1 (independent code paths).
4. **Tier 3** depends on E0.1 (reproduction on flat manifolds) and the matched-arm protocol; E3.2 depends on E0.2’s bias-extraction style.
5. **Tier 4** depends on E0.2 ( $S$ ) and E0.5 (rate, for the bias/variance split).

**Per-experiment validity-threat summary.** Each experiment’s “Validity safeguards” field is the controlling list. The recurring threats and their controls are: selection confounds (use singleton grids for any  $S$ -based or bias-isolation test); bias floors (use the optimal bandwidth schedule for rate tests); noise domination (use low  $\sigma$  for bias-isolation tests); averaging dilution (stratify by curvature/boundary); dependence leakage (grouped folds, disjoint test clusters); and metric/deploy mismatch (score the deployed quantity). A result whose safeguards were not met is reported as *inconclusive*, never as evidence.